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EXPERIMENT – 3

Aim : To deploy multi-service applications using Docker Compose

LO Mapped : LO4

LO : Containerize, orchestrate and deploy applications using Docker, Docker Compose, Kubernetes and cloud infrastructure tools.

Theory :

What is Docker Compose?

Docker Compose is a tool provided by Docker that is used to define, configure, and run multi-container applications.

In a typical application, different components may require different services. For example, a web application may need:

- A web server such as Nginx
- A database such as MySQL
- A cache such as Redis
- A database management tool such as phpMyAdmin

Instead of creating and managing each Docker container separately, Docker Compose allows us to define all these services in a single configuration file called:

docker-compose.yml

The configuration file is written in YAML (YAML Ain't Markup Language) format.

Once the services are defined in the YAML file, all the containers can be created and started together using a single command:

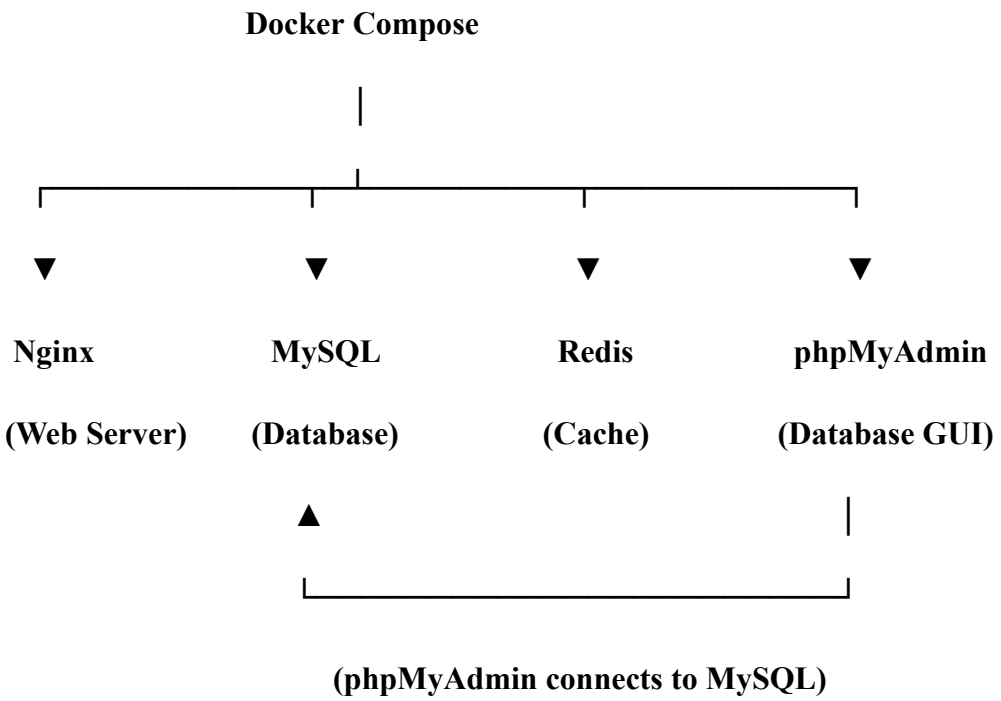
docker compose up -d

Here, -d means detached mode, which runs the containers in the background.

How Docker Compose Works

Docker Compose uses a YAML file to describe the application architecture.

For example:



Each component runs in its own Docker container, but Docker Compose manages them as a single application.

Docker Compose also creates a network for the services, allowing containers to communicate with each other using their service names.

For example:

phpMyAdmin ---> database

Here, database can be used as the hostname of the MySQL service inside the Compose network.

Docker Compose in Our Experiment

In our practical, Docker Compose was used to deploy a four-service application:

1. Nginx

Nginx was used as the web server.

It was accessed through:

http://localhost:8081

2. MySQL

MySQL was used as the database server.

It contained the database:

studentdb

3. Redis

Redis was used as an in-memory data store and cache.

It was tested using:

PING

The response:

PONG

confirmed that Redis was running.

4. phpMyAdmin

phpMyAdmin was used to manage the MySQL database through a web interface.

It was accessed through:

http://localhost:8082

Docker Compose Commands and Descriptions:

Command	**Description**
mkdir compose-demo	Creates a new project directory named compose-demo.
cd compose-demo	Enters the compose-demo directory.
notepad docker-compose.yml	Creates and opens the Docker Compose YAML configuration file.
dir	Lists files and folders in the current directory.
docker compose config	Validates the docker-compose.yml configuration.
docker compose up -d	Creates and starts all services in detached/background mode.
docker compose ps	Shows the status of Compose services/containers.
docker compose logs	Displays logs of all services.
docker compose logs -f	Continuously displays service logs.
docker compose stop	Stops running services without removing containers.
docker compose start	Starts previously stopped services.
docker compose restart	Restarts the services.
docker compose down	Stops and removes containers and the Compose network.
docker compose down -v	Removes containers, network, and volumes including stored database data.
docker compose images	Displays the Docker images used by the services.

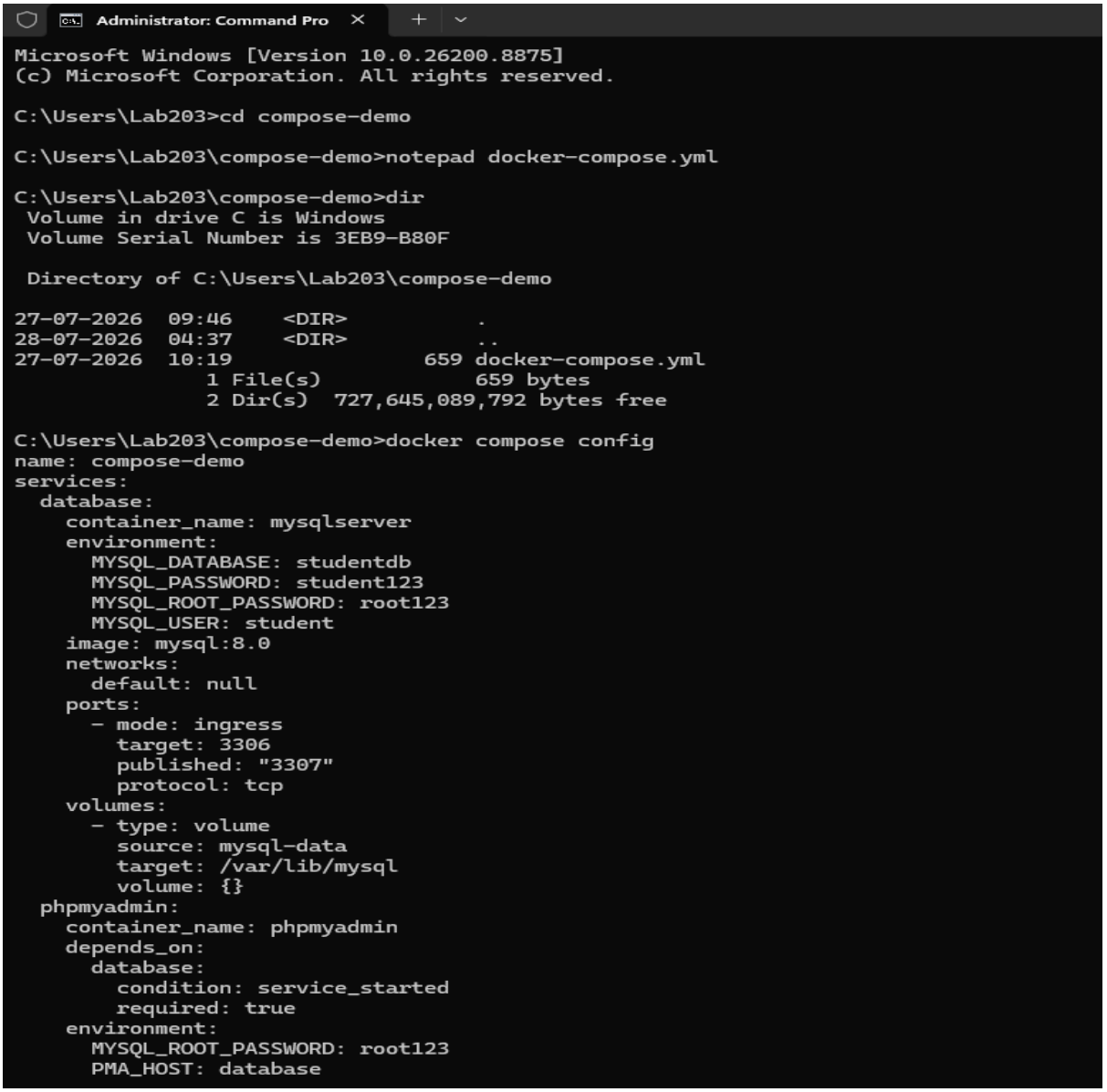
docker exec -it redisserver redis-cli	Opens the Redis command-line interface inside the Redis container.
docker ps	Shows currently running Docker containers.

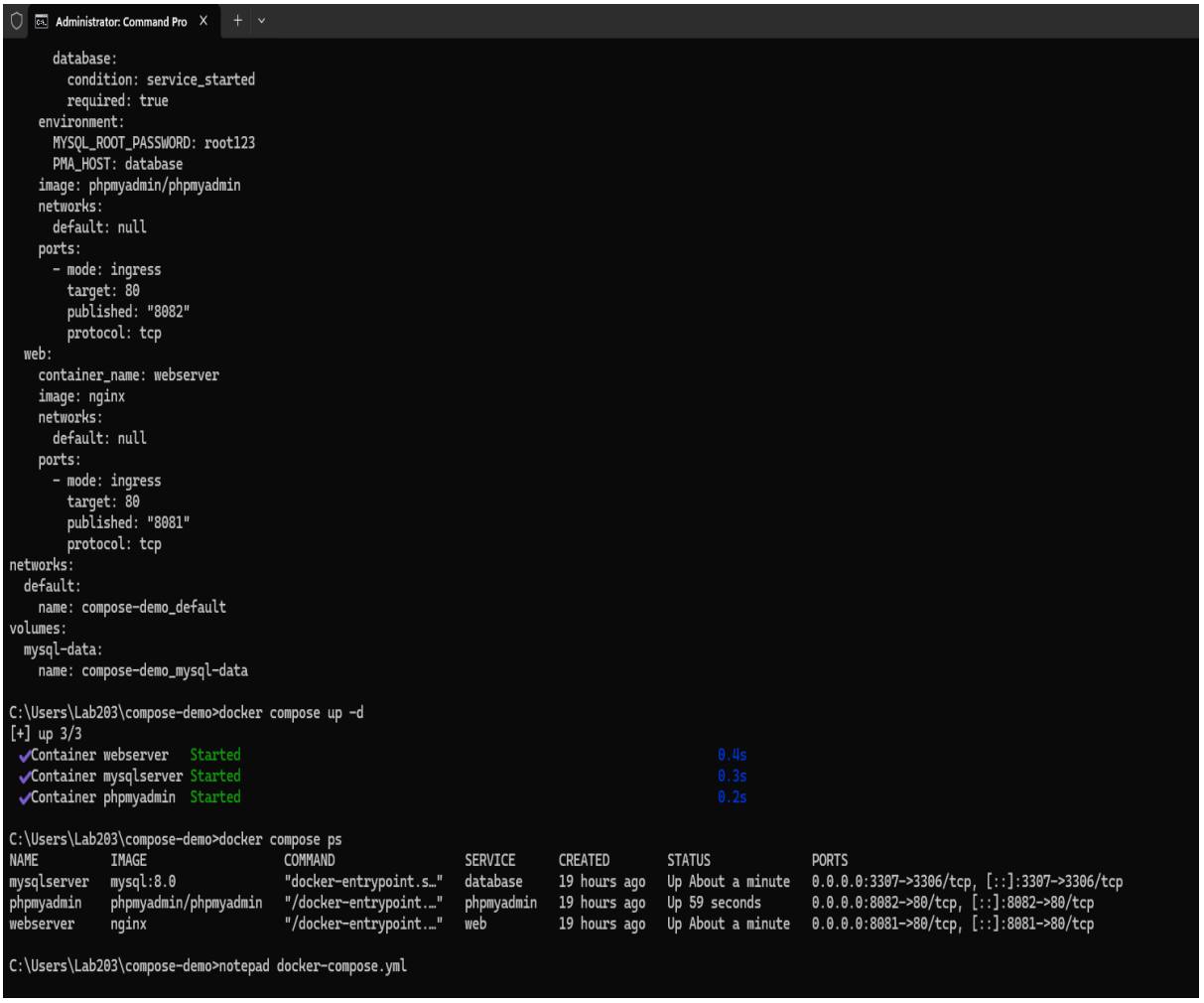
docker-compose.yml :

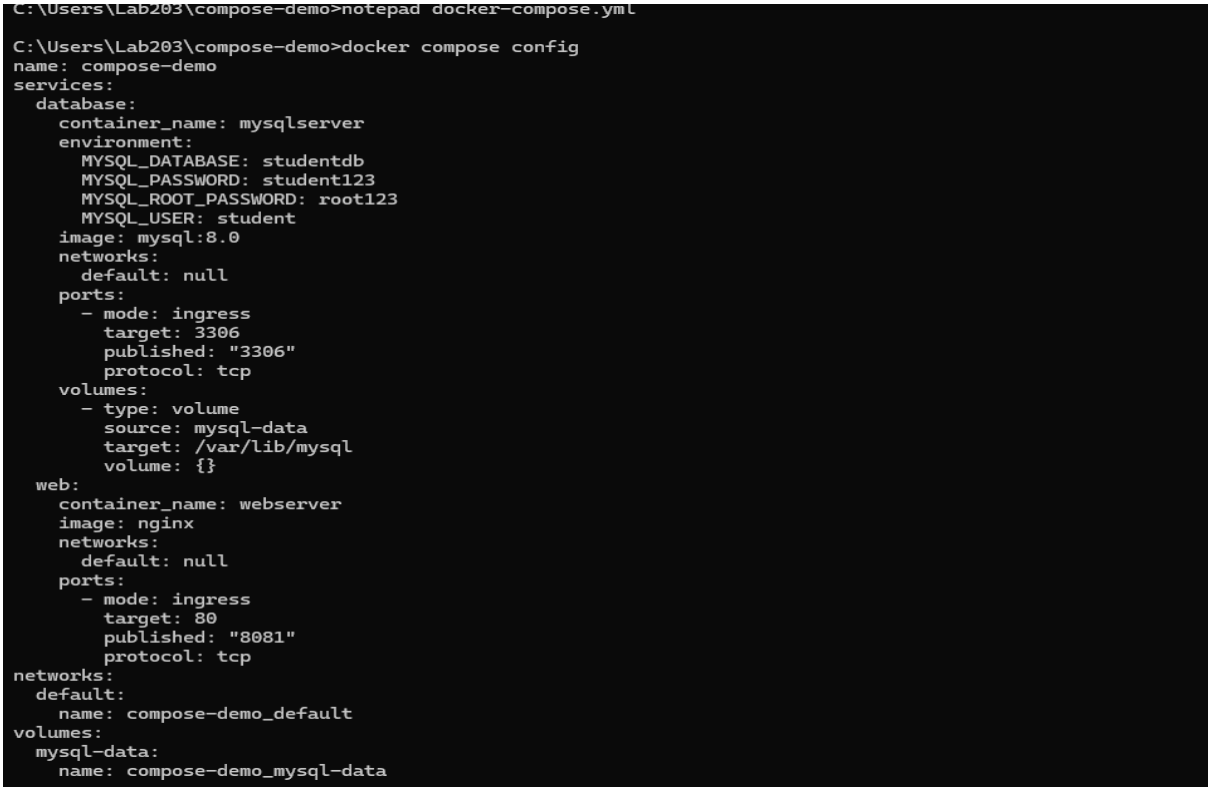
```
services:
  web:
    image: nginx
    container_name: webserver
    ports:
      - "8081:80"
  database:
    image: mysql:8.0
    container_name: mysqlserver
    environment:
      MYSQL_ROOT_PASSWORD: root123
      MYSQL_DATABASE: studentdb
      MYSQL_USER: student
      MYSQL_PASSWORD: student123
    ports:
      - "3307:3306"
    volumes:
```

```
    - mysql-data:/var/lib/mysql
  redis:
    image: redis:latest
    container_name: redisserver
    ports:
      - "6379:6379"
  phpmyadmin:
    image: phpmyadmin:latest
    container_name: phpmyadmin
    environment:
      PMA_HOST: database
      PMA_PORT: 3306
      PMA_USER: student
      PMA_PASSWORD: student123
    ports:
      - "8082:80"
    depends_on:
      - database
volumes:
  mysql-data:
```

Execution :


Administrator: Command Pro
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Lab203>cd compose-demo
C:\Users\Lab203\compose-demo>notepad docker-compose.yml
C:\Users\Lab203\compose-demo>dir
Volume in drive C is Windows
Volume Serial Number is 3EB9-B80F
Directory of C:\Users\Lab203\compose-demo
27-07-2026 09:46 <DIR> .
28-07-2026 04:37 <DIR> ..
27-07-2026 10:19 659 docker-compose.yml
1 File(s) 659 bytes
2 Dir(s) 727,645,089,792 bytes free
C:\Users\Lab203\compose-demo>docker compose config
name: compose-demo
services:
database:
container_name: mysqlserver
environment:
MYSQL_DATABASE: studentdb
MYSQL_PASSWORD: student123
MYSQL_ROOT_PASSWORD: root123
MYSQL_USER: student
image: mysql:8.0
networks:
default: null
ports:
- mode: ingress
target: 3306
published: "3307"
protocol: tcp
volumes:
- type: volume
source: mysql-data
target: /var/lib/mysql
volume: {}
phpmyadmin:
container_name: phpmyadmin
depends_on:
database:
condition: service_started
required: true
environment:
MYSQL_ROOT_PASSWORD: root123
PMA_HOST: database


Administrator: Command Pro
database:
condition: service_started
required: true
environment:
MYSQL_ROOT_PASSWORD: root123
PMA_HOST: database
image: phpmyadmin/phpmyadmin
networks:
default: null
ports:
- mode: ingress
target: 80
published: "8082"
protocol: tcp
web:
container_name: webserver
image: nginx
networks:
default: null
ports:
- mode: ingress
target: 80
published: "8081"
protocol: tcp
networks:
default:
name: compose-demo_default
volumes:
mysql-data:
name: compose-demo_mysql-data
C:\Users\Lab203\compose-demo>docker compose up -d
[+] up 3/3
Container webserver Started 0.4s
Container mysqlserver Started 0.3s
Container phpmyadmin Started 0.2s
C:\Users\Lab203\compose-demo>docker compose ps
NAME IMAGE COMMAND SERVICE CREATED STATUS PORTS
mysqlserver mysql:8.0 "docker-entrypoint.s…" database 19 hours ago Up About a minute 0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp
phpmyadmin phpmyadmin/phpmyadmin "/docker-entrypoint.…" phpmyadmin 19 hours ago Up 59 seconds 0.0.0.0:8082->80/tcp, [::]:8082->80/tcp
webserver nginx "/docker-entrypoint.…" web 19 hours ago Up About a minute 0.0.0.0:8081->80/tcp, [::]:8081->80/tcp
C:\Users\Lab203\compose-demo>notepad docker-compose.yml


C:\Users\Lab203\compose-demo>notepad docker-compose.yml
C:\Users\Lab203\compose-demo>docker compose config
name: compose-demo
services:
database:
container_name: mysqlserver
environment:
MYSQL_DATABASE: studentdb
MYSQL_PASSWORD: student123
MYSQL_ROOT_PASSWORD: root123
MYSQL_USER: student
image: mysql:8.0
networks:
default: null
ports:
- mode: ingress
target: 3306
published: "3306"
protocol: tcp
volumes:
- type: volume
source: mysql-data
target: /var/lib/mysql
volume: {}
web:
container_name: webserver
image: nginx
networks:
default: null
ports:
- mode: ingress
target: 80
published: "8081"
protocol: tcp
networks:
default:
name: compose-demo_default
volumes:
mysql-data:
name: compose-demo_mysql-data

```
C:\Users\Lab203\compose-demo>docker compose up -d
time="2026-07-28T05:17:53+05:30" level=warning msg="Found orphan containers (phpmyadmin) for this project. If you removed or renamed this service in your compose file, you can run this command with the --remov
e-orphans flag to clean it up."
[+] up 1/2
 ✔Container webserver   Running                                                                    0.0s
 - Container mysqlserver Starting                                                                  1.3s
Error response from daemon: ports are not available: exposing port TCP 0.0.0.0:3306 -> 127.0.0.1:0: listen tcp 0.0.0.0:3306: bind: Only one usage of each socket address (protocol/network address/port) is norma
lly permitted.

What's next:
    Debug this Compose error with Gordon → docker ai "help me fix this compose error"

C:\Users\Lab203\compose-demo>notepad docker-compose.yml

C:\Users\Lab203\compose-demo>docker compose down
[+] down 3/3
 ✔Container mysqlserver       Removed                                                              0.0s
 ✔Container webserver         Removed                                                              0.6s
 ! Network compose-demo_default Resource is still in use                                           0.0s

C:\Users\Lab203\compose-demo>docker compose up -d
time="2026-07-28T05:19:20+05:30" level=warning msg="Found orphan containers (phpmyadmin) for this project. If you removed or renamed this service in your compose file, you can run this command with the --remov
e-orphans flag to clean it up."
[+] up 2/2
 ✔Container webserver   Started                                                                    0.3s
 ✔Container mysqlserver Started                                                                    0.2s
```

```
C:\Users\Lab203\compose-demo>docker compose logs database
mysqlserver  | 2026-07-27 23:49:20+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.46-1.el9 started.
mysqlserver  | 2026-07-27 23:49:20+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
mysqlserver  | 2026-07-27 23:49:20+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.46-1.el9 started.
mysqlserver  | '/var/lib/mysql/mysql.sock' -> '/var/run/mysqld/mysqld.sock'
mysqlserver  | 2026-07-27T23:49:20.663745Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future release. Please use SET GLOBAL host_cache_size=0 instea
d.
mysqlserver  | 2026-07-27T23:49:20.664777Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.0.46) starting as process 1
mysqlserver  | 2026-07-27T23:49:20.678486Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
mysqlserver  | 2026-07-27T23:49:20.803017Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
mysqlserver  | 2026-07-27T23:49:20.964728Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
mysqlserver  | 2026-07-27T23:49:20.964811Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this channel.
mysqlserver  | 2026-07-27T23:49:20.966955Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choosing a dif
erent directory.
mysqlserver  | 2026-07-27T23:49:20.981965Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var/run/mysqld/mysqlx.sock
mysqlserver  | 2026-07-27T23:49:20.982043Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.0.46'  socket: '/var/run/mysqld/mysqld.sock'  port: 3306  MySQL Community Server
- GPL.

C:\Users\Lab203\compose-demo>docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED             STATUS             PORTS                                                  NAMES
5f61cbdc0ad5   nginx                  "/docker-entrypoint.…"   About a minute ago  Up About a minute  0.0.0.0:8081->80/tcp, [::]:8081->80/tcp                webserver
4911ad7b7122   mysql:8.0              "docker-entrypoint.s…"   About a minute ago  Up About a minute  0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp             mysqlserver
a0842d265f53   phpmyadmin/phpmyadmin  "/docker-entrypoint.…"   19 hours ago        Up 6 minutes       0.0.0.0:8082->80/tcp, [::]:8082->80/tcp                phpmyadmin
```

```
C:\Users\Lab203\compose-demo>docker exec -it mysqlserver mysql -u student -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.46 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| performance_schema |
| studentdb          |
+--------------------+
3 rows in set (0.02 sec)

mysql> exit
Bye
```

```
C:\Users\Lab203\compose-demo>docker compose logs
mysqlserver  | 2026-07-27 23:49:20+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.46-1.el9 started.
mysqlserver  | 2026-07-27 23:49:20+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
mysqlserver  | 2026-07-27 23:49:20+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.46-1.el9 started.
mysqlserver  | '/var/lib/mysql/mysql.sock' -> '/var/run/mysqld/mysqld.sock'
mysqlserver  | 2026-07-27T23:49:20.663745Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future release. Please use SET GLOBAL host_cache_size=0 instead
.
mysqlserver  | 2026-07-27T23:49:20.664777Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.0.46) starting as process 1
mysqlserver  | 2026-07-27T23:49:20.678486Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
mysqlserver  | 2026-07-27T23:49:20.803017Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
mysqlserver  | 2026-07-27T23:49:20.964728Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
mysqlserver  | 2026-07-27T23:49:20.964811Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this channel.
webserver    | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
webserver    | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
webserver    | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
webserver    | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
mysqlserver  | 2026-07-27T23:49:20.966955Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choosing a diff
erent directory.
mysqlserver  | 2026-07-27T23:49:20.981965Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var/run/mysqld/mysqlx.sock
mysqlserver  | 2026-07-27T23:49:20.982043Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.0.46'  socket: '/var/run/mysqld/mysqld.sock'  port: 3306  MySQL Community Server
- GPL.
webserver    | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
webserver    | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
webserver    | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
webserver    | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
webserver    | /docker-entrypoint.sh: Configuration complete; ready for start up
webserver    | 2026/07/27 23:49:20 [notice] 1#1: using the "epoll" event method
webserver    | 2026/07/27 23:49:20 [notice] 1#1: nginx/1.31.3
webserver    | 2026/07/27 23:49:20 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
webserver    | 2026/07/27 23:49:20 [notice] 1#1: OS: Linux 5.15.167.4-microsoft-standard-WSL2
webserver    | 2026/07/27 23:49:20 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
webserver    | 2026/07/27 23:49:20 [notice] 1#1: start worker processes
webserver    | 2026/07/27 23:49:20 [notice] 1#1: start worker process 29
webserver    | 2026/07/27 23:49:20 [notice] 1#1: start worker process 30
webserver    | 2026/07/27 23:49:20 [notice] 1#1: start worker process 31
webserver    | 2026/07/27 23:49:20 [notice] 1#1: start worker process 32
webserver    | 2026/07/27 23:49:20 [notice] 1#1: start worker process 33
webserver    | 2026/07/27 23:49:20 [notice] 1#1: start worker process 34
webserver    | 2026/07/27 23:49:20 [notice] 1#1: start worker process 35
webserver    | 2026/07/27 23:49:20 [notice] 1#1: start worker process 36
webserver    | 2026/07/27 23:49:20 [notice] 1#1: start worker process 37
webserver    | 2026/07/27 23:49:20 [notice] 1#1: start worker process 38
webserver    | 2026/07/27 23:49:20 [notice] 1#1: start worker process 39
webserver    | 2026/07/27 23:49:20 [notice] 1#1: start worker process 40
webserver    | 2026/07/27 23:49:20 [notice] 1#1: start worker process 41
webserver    | 2026/07/27 23:49:20 [notice] 1#1: start worker process 42
webserver    | 2026/07/27 23:49:20 [notice] 1#1: start worker process 43
webserver    | 2026/07/27 23:49:20 [notice] 1#1: start worker process 44
webserver    | 2026/07/27 23:49:20 [notice] 1#1: start worker process 45
```

```
C:\Users\Lab203\compose-demo>docker compose stop
[+] stop 2/2
 ✔Container mysqlserver Stopped                                                    1.3s
 ✔Container webserver   Stopped                                                    0.6s

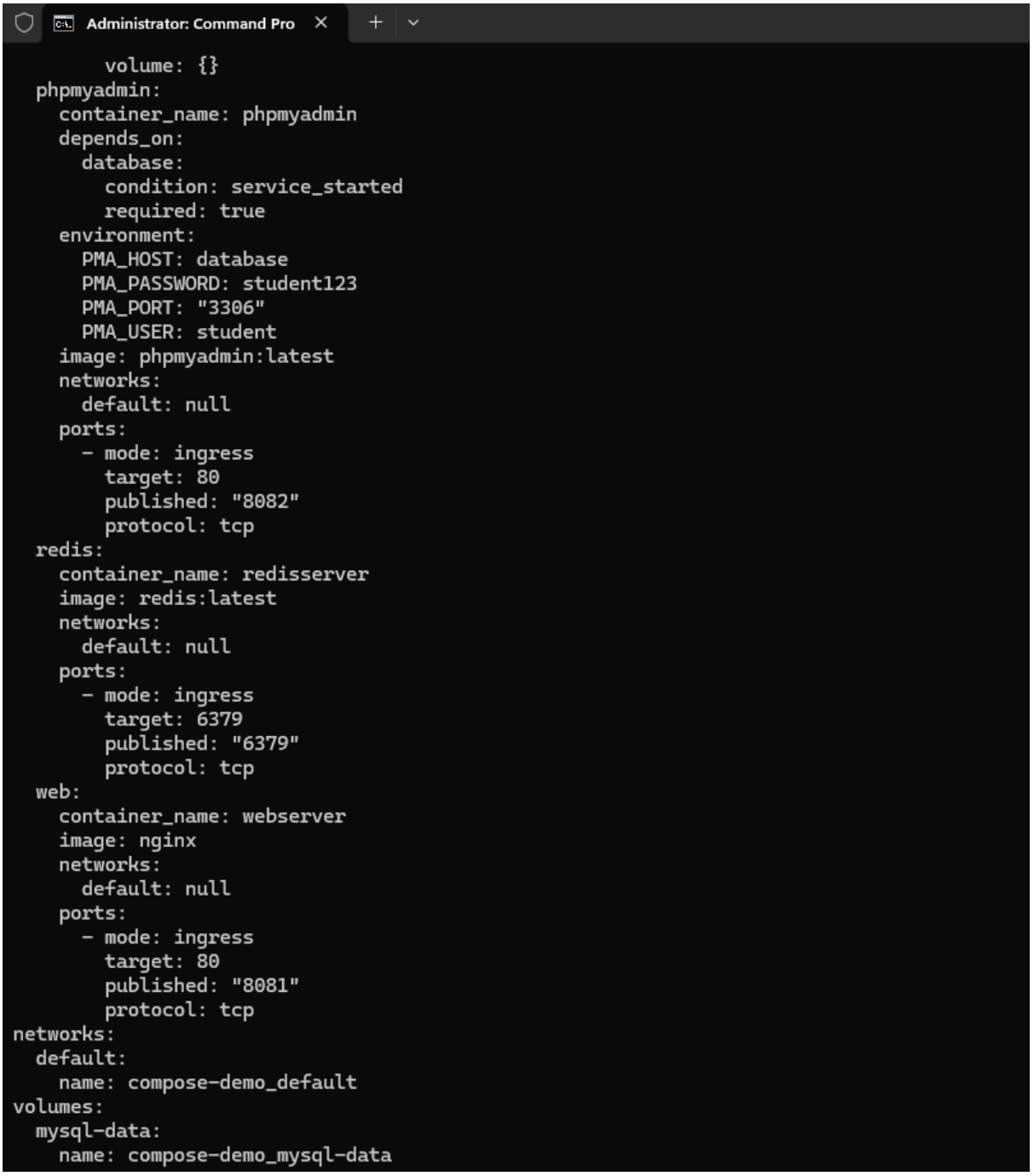
C:\Users\Lab203\compose-demo>docker compose start
[+] start 2/2
 ✔Container mysqlserver Started                                                    0.2s
 ✔Container webserver   Started                                                    0.2s

C:\Users\Lab203\compose-demo>docker compose restart
[+] restart 0/2
 - Container mysqlserver Restarting                                                1.8s
 - Container webserver   Restarting                                                1.8s

C:\Users\Lab203\compose-demo>docker compose down
[+] down 3/3
 ✔Container mysqlserver         Removed                                           1.8s
 ✔Container webserver           Removed                                           0.6s
 ! Network compose-demo_default Resource is still in use                          0.0s

C:\Users\Lab203\compose-demo>notepad docker-compose.yml

C:\Users\Lab203\compose-demo>docker compose config
name: compose-demo
services:
  database:
    container_name: mysqlserver
    environment:
      MYSQL_DATABASE: studentdb
      MYSQL_PASSWORD: student123
      MYSQL_ROOT_PASSWORD: root123
      MYSQL_USER: student
    image: mysql:8.0
    networks:
      default: null
    ports:
      - mode: ingress
        target: 3306
        published: "3307"
        protocol: tcp
    volumes:
      - type: volume
        source: mysql-data
        target: /var/lib/mysql
        volume: {}
```


Administrator: Command Pro
volume: {}
phpmyadmin:
container_name: phpmyadmin
depends_on:
database:
condition: service_started
required: true
environment:
PMA_HOST: database
PMA_PASSWORD: student123
PMA_PORT: "3306"
PMA_USER: student
image: phpmyadmin:latest
networks:
default: null
ports:
- mode: ingress
target: 80
published: "8082"
protocol: tcp
redis:
container_name: redisserver
image: redis:latest
networks:
default: null
ports:
- mode: ingress
target: 6379
published: "6379"
protocol: tcp
web:
container_name: webserver
image: nginx
networks:
default: null
ports:
- mode: ingress
target: 80
published: "8081"
protocol: tcp
networks:
default:
name: compose-demo_default
volumes:
mysql-data:
name: compose-demo_mysql-data

```
C:\Users\Lab203\compose-demo>docker compose down --remove-orphans
[+] down 2/2
 ✔Container phpmyadmin         Removed                                                    1.5s
 ✔Network compose-demo_default Removed                                                    0.5s

C:\Users\Lab203\compose-demo>docker compose up -d
[+] up 35/35
 ✔Image phpmyadmin:latest      Pulled                                                    17.9s
 ✔Image redis:latest           Pulled                                                     8.2s
 ✔Network compose-demo_default Created                                                    0.0s
 ✔Container webserver          Started                                                    0.7s
 ✔Container mysqlserver        Started                                                    0.8s
 ✔Container redisserver        Started                                                    0.8s
 ✔Container phpmyadmin         Started                                                    0.5s

C:\Users\Lab203\compose-demo>docker compose ps
NAME          IMAGE               COMMAND                  SERVICE      CREATED          STATUS          PORTS
mysqlserver   mysql:8.0           "docker-entrypoint.s…"   database     15 seconds ago   Up 13 seconds   0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp
phpmyadmin    phpmyadmin:latest   "/docker-entrypoint.…"   phpmyadmin   14 seconds ago   Up 13 seconds   0.0.0.0:8082->80/tcp, [::]:8082->80/tcp
redisserver   redis:latest        "docker-entrypoint.s…"   redis        15 seconds ago   Up 13 seconds   0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp
webserver     nginx               "/docker-entrypoint.…"   web          15 seconds ago   Up 13 seconds   0.0.0.0:8081->80/tcp, [::]:8081->80/tcp

C:\Users\Lab203\compose-demo>docker exec -it mysqlserver mysql -u student -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 20
Server version: 8.0.46 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| performance_schema |
| studentdb          |
+--------------------+
```

```
Administrator: Command Pro

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| performance_schema |
| studentdb          |
+--------------------+
3 rows in set (0.00 sec)

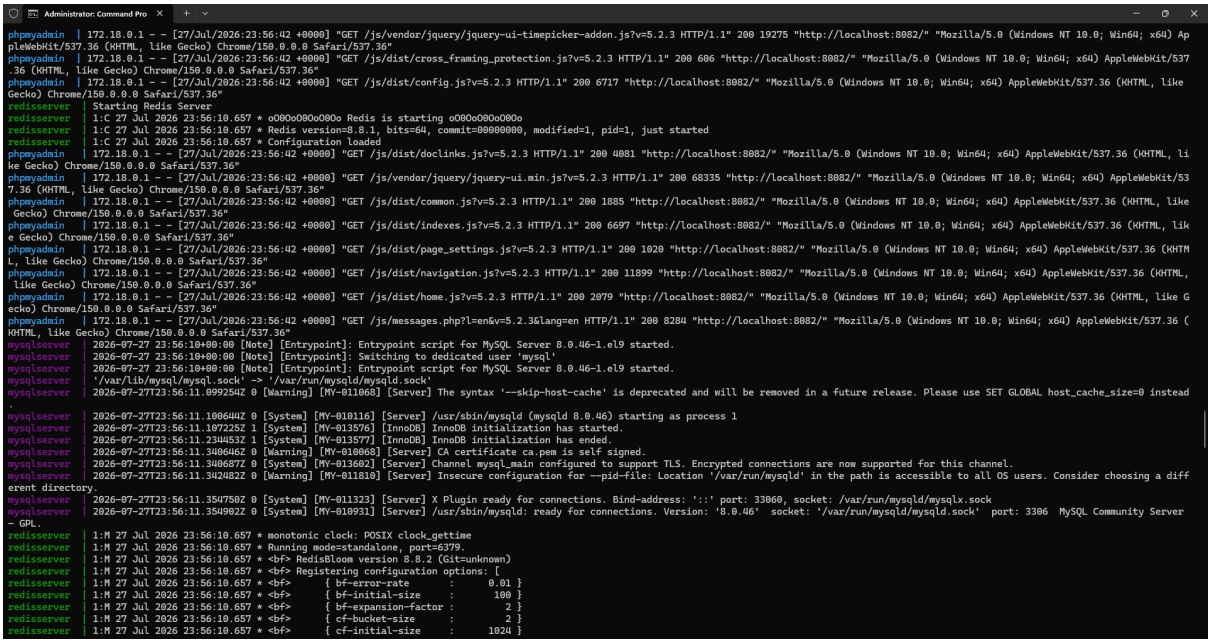
mysql> exit
Bye

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug mysqlserver
    Learn more at https://docs.docker.com/go/debug-cli/

C:\Users\Lab203\compose-demo>docker exec -it redisserver redis-cli
127.0.0.1:6379> ping
PONG
127.0.0.1:6379> exit

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug redisserver
    Learn more at https://docs.docker.com/go/debug-cli/

C:\Users\Lab203\compose-demo>docker compose logs
phpmyadmin  | AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.18.0.5. Set the 'ServerName' directive globally to suppress this message
phpmyadmin  | AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.18.0.5. Set the 'ServerName' directive globally to suppress this message
phpmyadmin  | [Mon Jul 27 23:56:10.913262 2026] [mpm_prefork:notice] [pid 1:tid 1] AH00163: Apache/2.4.68 (Debian) PHP/8.3.32 configured -- resuming normal operations
phpmyadmin  | [Mon Jul 27 23:56:10.913293 2026] [core:notice] [pid 1:tid 1] AH00094: Command line: 'apache2 -D FOREGROUND'
phpmyadmin  | 172.18.0.1 - - [27/Jul/2026:23:56:42 +0000] "GET / HTTP/1.1" 200 18368 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin  | 172.18.0.1 - - [27/Jul/2026:23:56:42 +0000] "GET /themes/pmahomme/jquery/jquery-ui.css HTTP/1.1" 200 8823 "http://localhost:8082/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin  | 172.18.0.1 - - [27/Jul/2026:23:56:42 +0000] "GET /js/vendor/codemirror/lib/codemirror.css?v=5.2.3 HTTP/1.1" 200 2848 "http://localhost:8082/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin  | 172.18.0.1 - - [27/Jul/2026:23:56:42 +0000] "GET /js/vendor/codemirror/addon/hint/show-hint.css?v=5.2.3 HTTP/1.1" 200 668 "http://localhost:8082/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin  | 172.18.0.1 - - [27/Jul/2026:23:56:42 +0000] "GET /js/vendor/codemirror/addon/lint/lint.css?v=5.2.3 HTTP/1.1" 200 1634 "http://localhost:8082/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin  | 172.18.0.1 - - [27/Jul/2026:23:56:42 +0000] "GET /js/vendor/jquery/jquery-migrate.min.js?v=5.2.3 HTTP/1.1" 200 5401 "http://localhost:8082/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin  | 172.18.0.1 - - [27/Jul/2026:23:56:42 +0000] "GET /js/vendor/sprintf.js?v=5.2.3 HTTP/1.1" 200 2970 "http://localhost:8082/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin  | 172.18.0.1 - - [27/Jul/2026:23:56:42 +0000] "GET /js/dist/keyhandler.js?v=5.2.3 HTTP/1.1" 200 1116 "http://localhost:8082/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
phpmyadmin  | 172.18.0.1 - - [27/Jul/2026:23:56:42 +0000] "GET /js/dist/ajax.js?v=5.2.3 HTTP/1.1" 200 8188 "http://localhost:8082/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36"
```




```
C:\Users\Lab203\compose-demo>docker compose stop
[+] stop 4/4
 ✔Container phpmyadmin  Stopped                                    1.6s
 ✔Container webserver   Stopped                                    0.8s
 ✔Container redisserver Stopped                                    0.6s
 ✔Container mysqlserver Stopped                                    1.7s

C:\Users\Lab203\compose-demo>docker compose ps
NAME      IMAGE     COMMAND   SERVICE   CREATED   STATUS    PORTS

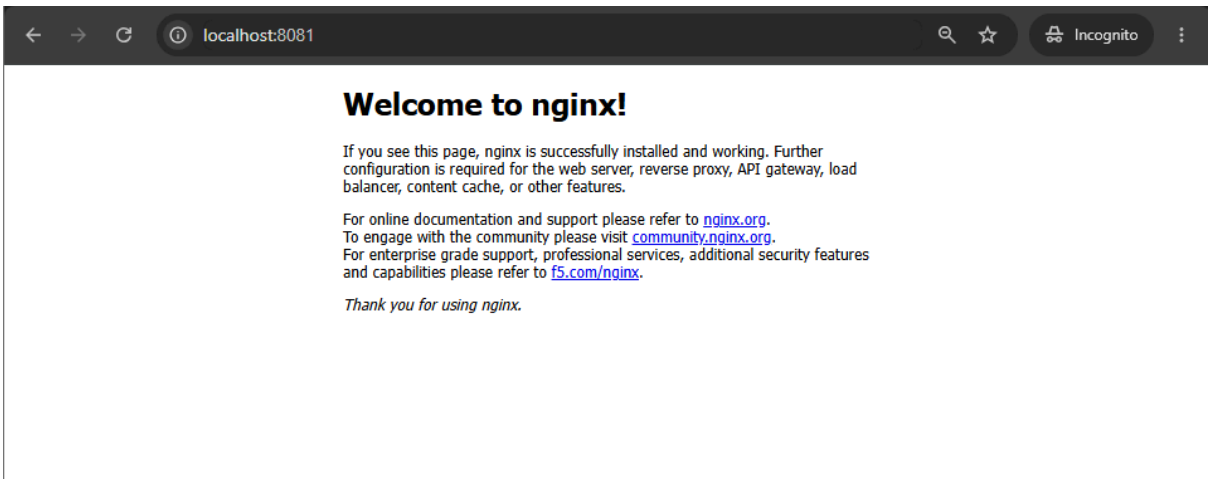
C:\Users\Lab203\compose-demo>docker compose start
[+] start 4/4
 ✔Container redisserver Started                                    0.3s
 ✔Container webserver   Started                                    0.2s
 ✔Container mysqlserver Started                                    0.2s
 ✔Container phpmyadmin  Started                                    0.2s

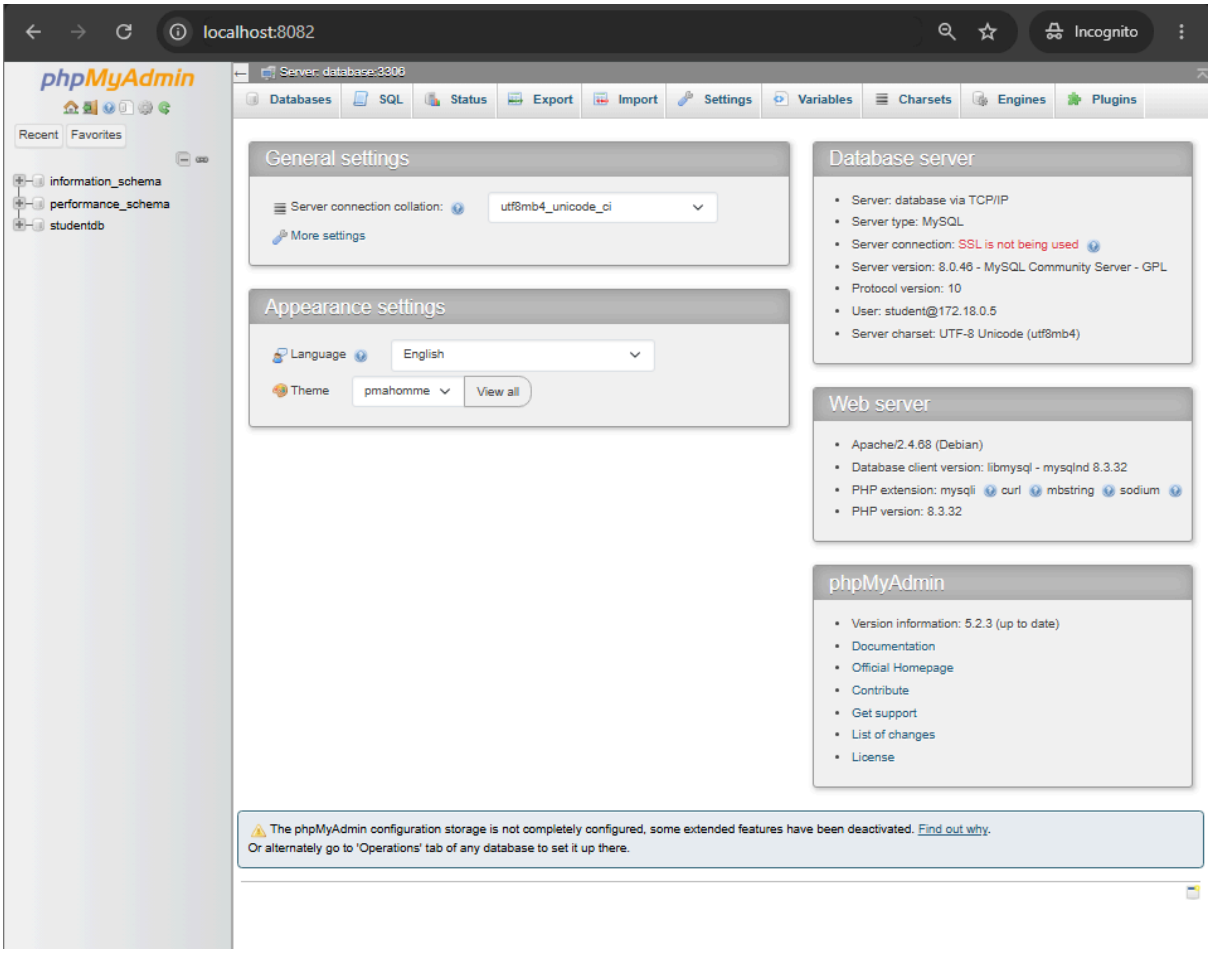
C:\Users\Lab203\compose-demo>docker compose ps
NAME          IMAGE                COMMAND                  SERVICE      CREATED         STATUS         PORTS
mysqlserver   mysql:8.0            "docker-entrypoint.s…"   database     3 minutes ago   Up 5 seconds   0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp
phpmyadmin    phpmyadmin:latest    "/docker-entrypoint.…"   phpmyadmin   3 minutes ago   Up 5 seconds   0.0.0.0:8082->80/tcp, [::]:8082->80/tcp
redisserver   redis:latest         "docker-entrypoint.s…"   redis        3 minutes ago   Up 5 seconds   0.0.0.0:6379->6379/tcp, [::]:6379->6379/tcp
webserver     nginx                "/docker-entrypoint.…"   web          3 minutes ago   Up 5 seconds   0.0.0.0:8081->80/tcp, [::]:8081->80/tcp
```

```
C:\Users\Lab203\compose-demo>docker compose restart
[+] restart 0/4
 - Container webserver   Restarting                                 2.3s
 - Container phpmyadmin  Restarting                                 2.3s
 - Container mysqlserver Restarting                                 2.3s
 - Container redisserver Restarting                                 2.3s

C:\Users\Lab203\compose-demo>docker compose down
[+] down 5/5
 ✔Container webserver          Removed                              0.9s
 ✔Container phpmyadmin         Removed                              1.5s
 ✔Container redisserver        Removed                              0.7s
 ✔Container mysqlserver        Removed                              2.1s
 ✔Network compose-demo_default Removed                              0.5s

C:\Users\Lab203\compose-demo>
```





Conclusion :

The multi-service application was successfully deployed using **Docker Compose**. Four services—**Nginx, MySQL, Redis, and phpMyAdmin**—were configured and managed using a docker-compose.yml file. The services were successfully started, tested, monitored, stopped, and restarted using Docker Compose commands.